Tom Reshef-Israeli

PhD candidate in physics and science education
Faculty of Education in Science and Technology, Technion Israel Institute of Technology
tomis@campus.technion.ac.il | tom.is0501@gmail.com
Google Scholar | ORCID | LinkedIn | ResearchGate | Website

Education

Technion Israel Institute of Technology 2023–present
PhD, Education in Technology and Science

- Research: Unpacking collaborative sensemaking and modeling in physics education.
- Advisor: Shulamit Kapon; expected thesis submission: September 2027.

The Hebrew University of Jerusalem 2019
MSc, Physics

- Advisor: Michael Assaf.
- Thesis: The effect of a time-varying environment on population switching.

David Yellin College 2019
Teaching Certificate, Secondary School Physics

The Hebrew University of Jerusalem 2017
BSc, Physics and Cognitive Science

Current research projects

Projects in which I currently participate:

- **Mathematical modeling in physics learning:** Examining modeling pathways, shared understanding, epistemic uncertainty, and teacher interventions.
- **Learner-level discovery through mathematics:** Conceptualizing, designing and investigating classroom activities.
- **Framing and learning in collaborative laboratory inquiry:** Analyzing the unfolding of students' framing alongside their instructor's.
- **Experimental practices:** Developing curriculum and classroom materials for secondary school physics laboratories.

Selected fellowships, awards and honors

Faculty Prize for Academic Excellence in Research, Technion 2026
APS GPER Travel Grant Award 2026
Miriam and Aaron Gutwirth PhD Excellence Scholarship, Technion 2025–2026
Graduate Research Day Poster Award, 2nd place, Technion 2025
David and Sylvia Fine PhD Excellence Scholarship, Technion 2024–2025
Certificate of Excellence in Teaching, Consistent Performance, Technion 2024–2025
PERTG and AAPT Travel Grant Award 2024
Nano Excellence Fellowship; Certificate of Excellence in Teaching Physics 2017–2019
The Hebrew University of Jerusalem; teaching excellence in two consecutive years.

Additional academic training

Visiting student, University of Maryland Winter 2025–2026
University of Maryland, College Park – Andrew Elby's group
Continued doctoral research and participated in research group meetings.

Doctoral summer school: Building Theory in Science Education Research August 2025
Malmö University, Sweden
2 ECTS.

Publications

Journal Articles

Dynamics of collaborative modeling in an ill-structured real-world problem 2026
T. Reshef-Israeli and S. Kapon.
Physical Review Physics Education Research, 22(1), 010141. Editors' Suggestion. [DOI]

From mathematical structure to physical discovery: A curriculum-grounded characterization In press
T. Reshef-Israeli and D. Klein.
Physical Review Physics Education Research. [DOI]

Measuring air density with a falling beach ball In press
T. Reshef-Israeli and S. Kapon.
The Physics Teacher.

Layered modeling of a real-world drag-force problem: A pedagogical journey from idealized to realistic physics 2025
T. Reshef-Israeli and S. Kapon.
European Journal of Physics, 46(6), 065701. [DOI]

Engaging students in collaborative layered modeling of a real-world problem 2025
T. Reshef-Israeli and S. Kapon.
Physics Education, 60(5), 055026. [DOI]

Population switching under a time-varying environment 2020
T. Israeli and M. Assaf.
Physical Review E, 101(2), 022109. [DOI]

Under Review

Tracing epistemic uncertainty in real time: Functions and peer uptake in collaborative mathematical modelling Under review
T. Reshef-Israeli and S. Kapon.
ZDM – Mathematics Education.

Refereed Conference Proceedings

Collaborative sensemaking: The emergence of insights in a distributed knowledge system 2024
T. Reshef-Israeli and S. Kapon.
Physics Education Research Conference Proceedings, 361–366. [DOI]

Teacher Journals (Hebrew)

From following instructions to developing experimental thinking: The case of capacitor charging 2026
T. Reshef-Israeli.
Tehuda – The Journal of the Israeli Physics Teachers, 42, 43–58. [DOI]

Dialogue in the dark: The curious incident of ultraviolet in the night-time 2025
T. Reshef-Israeli.
Tehuda – The Journal of the Israeli Physics Teachers, 41, 5–8. [DOI]

Energy conservation – without energy! 2022
T. Reshef-Israeli.
Tehuda – The Journal of the Israeli Physics Teachers, 39(1–2), 4–16. [DOI]

Parachuting – from Leonardo da Vinci to the Millikan experiment 2020
T. Israeli.
Tehuda – The Journal of the Israeli Physics Teachers, 38(1–2), 55–76. [DOI]

Teaching and teacher education

ACADEMIC AND SCHOOL TEACHING

Technion – Teaching assistant 2022–present
Methods of Teaching Physics 1 (2022–present) and 2 (2023–present); Physics Lab 1m (2023–present) and 2m (2024–present).

The Hebrew University of Jerusalem – Teaching assistant 2017–2019
Electricity and Magnetism (2017, 2018); Physics Lab III: advanced experiments in quantum and modern physics (2018, 2019).

Physics and mathematics teacher, Leo Baeck Education Center, Haifa 2019–2023

Physics teacher, Sieff & Marks School, Jerusalem 2016–2018

TEACHER EDUCATION AND CURRICULUM DEVELOPMENT

Scientific experimentation learning materials, Technion 2026–present
Developing secondary school teaching sequences for modeling physical phenomena.

Professional development for in-service physics teachers, Technion 2025–2026
Leading courses on higher-order physical reasoning (2026) and modeling through kinematics (2025, 2026).

Online course on the physics of motion, Technion 2022–2024
Script writer and filmed teacher for a MOOC developing modeling competencies through kinematics in grades 9 and 10.

Physics teacher courses and conference workshops 2021–2023
Led four Ministry of Education courses on integrating physics MOOCs; organized and presented workshops at the Israeli Physics Teachers Conference, Weizmann Institute of Science.

Selected conference presentations

ORAL PRESENTATIONS

Coauthored with Shulamit Kapon.

Epistemic uncertainty across mathematical modelling in collaborative physics problem-solving 2026
GIREP–MPTL, Rethymno, Greece – Symposium presentation

Negotiating social and epistemic participation in undergraduate physics group work 2026
NARST – Online oral presentation

Unpacking the process of mathematical modeling in collaborative problem-solving 2026
APS Global Physics Summit, Denver, CO

The evolution of shared knowledge in collaborative problem-solving 2025
ESERA, Copenhagen, Denmark

Epistemic and social features of the emergence of insights in collaborative sensemaking 2024
Physics Education Research Conference, Boston, MA – Symposium presentation

POSTER PRESENTATIONS

Visualizing shared knowledge evolution in collaborative problem-solving 2025
Faculty and Jacobs Graduate Research Days, Technion – with S. Kapon

The effect of environmental variations on switching of metastable populations 2019
Statistical Physics of Complex Systems, Nordita, Stockholm – with M. Assaf

Academic service and memberships

Journal and proceedings reviewing: Physical Review Physics Education Research (2026); European Journal of Physics (2026); Science & Education (2025–2026); Physics Education Research Conference Proceedings (2024).

Conference reviewing: NARST (2026, 2027); GIREP–MPTL (2026); ESERA (2025).

Memberships: APS (2025–2027); AAPT (2024–2025); ESERA (2024, 2025); ESERA SIG8, FUTURES-oriented Science Education, and SIG12, Systems Thinking and Modeling in Science Education (2025–2026).